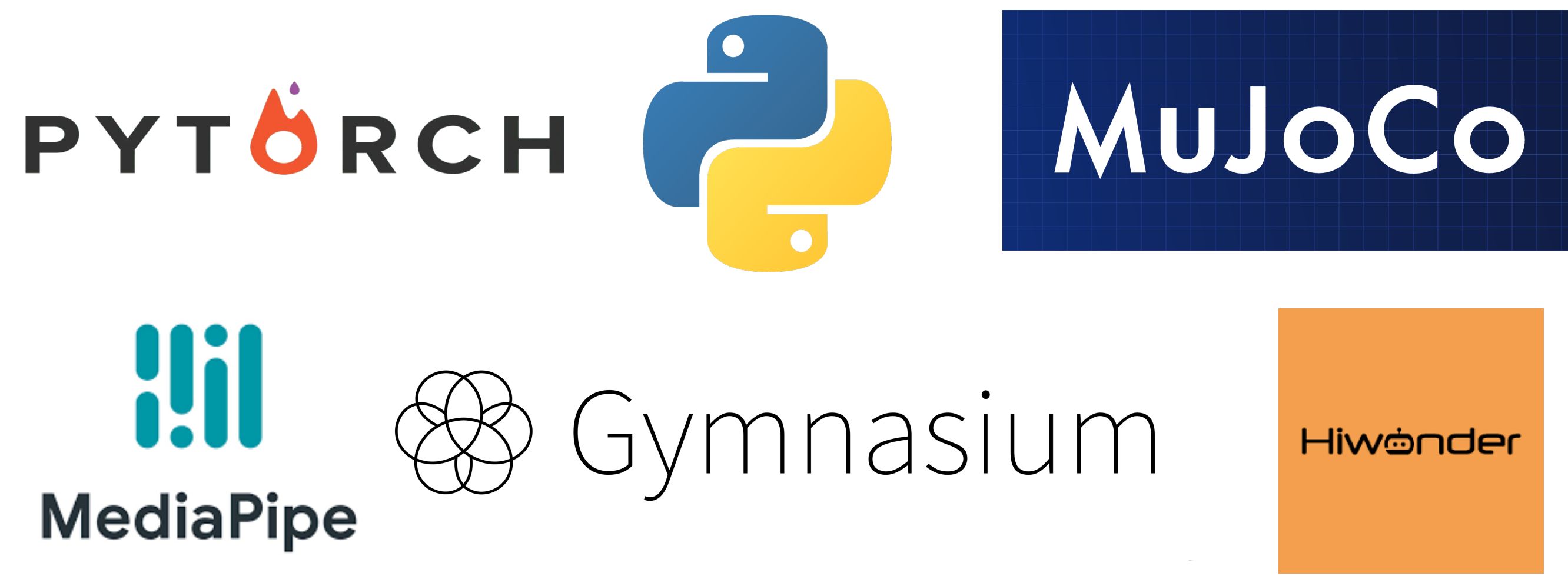


**Team:** Taylor Davis, Victor Rodriguez, Clayton Ramsey, Lucas Larson **Client:** Zach Lerner, PhD, Carlo R. da Cunha, PhD

## What Problem Are We Solving?

- **Rehabilitation robotics** relies on safe and natural physical interactions
- Traditional control methods lack adaptability and perception of human intent
- REACH explores **reinforcement learning (RL)** and **Gesture Recognition** to teach robots cooperative actions for **stroke victim rehabilitation**

## Technology Used



## Our Approach

- We have trained a **simulated robotic arm** in a physics-accurate environment called **MuJoCo**
- A **PPO** reinforcement learning agent learns **adaptive** control policies
- Policies are saved as compressed files and can be **replayed or transferred** to hardware
- A **MediaPipe CNN** recognizes hand gestures and inference is controlled via the onboard Raspberry Pi
- The AINex's ROS is Noetic and the bridge is built into the ROS Topic API

## Implemented Features

- **Simulation Environment** – With Python's implementation of MuJoCo, we can accurately simulate physics and robotic kinematics
- **Gesture Recognition** – Using Google's MediaPipe package, we can recognize various hand gestures
- **Framework Adaptability** – YAML-driven simulations share one RL train & evaluation workflow across arms, scenes, local runs, and cluster jobs
- **ROS Integration** – Through the use of the **ROS Topic** interface, we can communicate with the hardware, run policies, and log output

## System Overview

